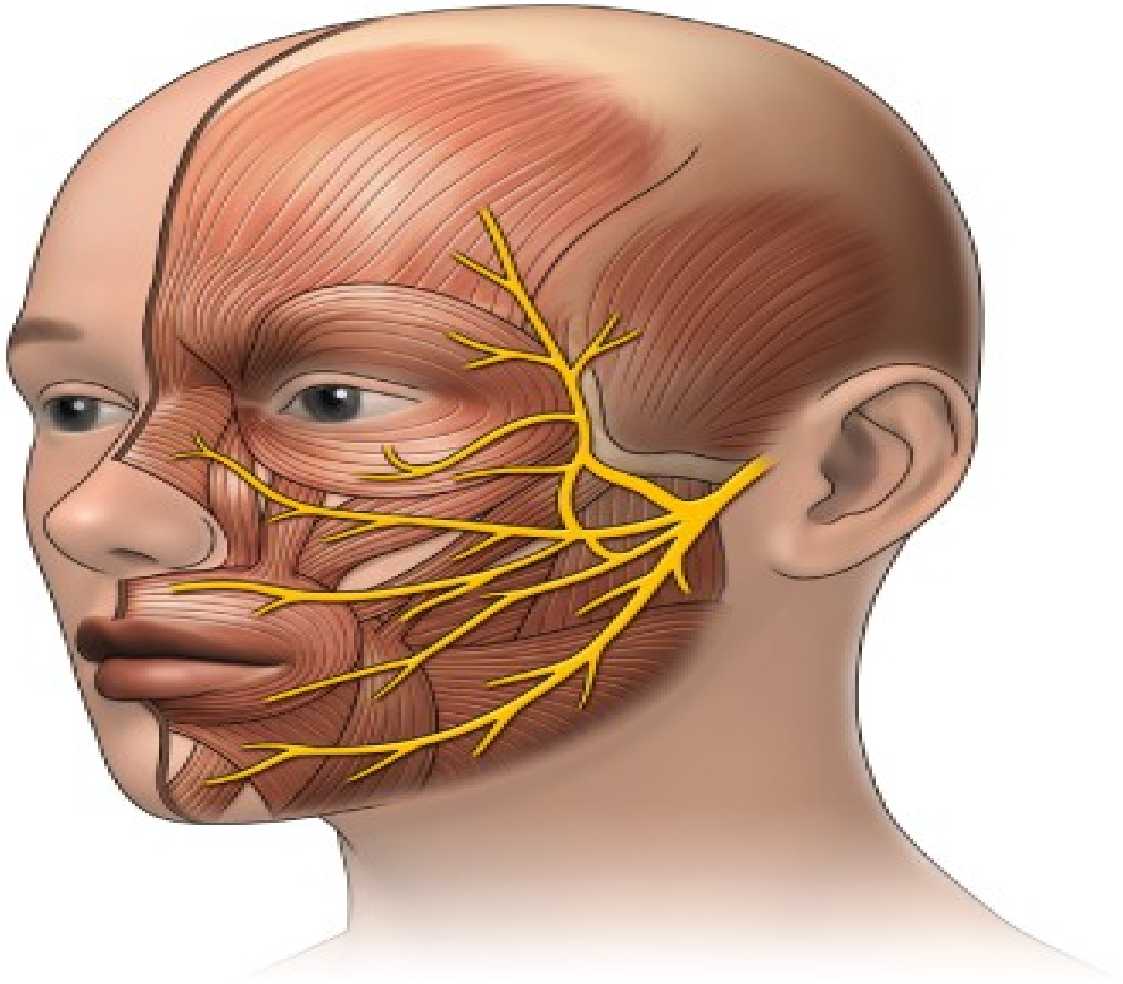
The Facial Nerve



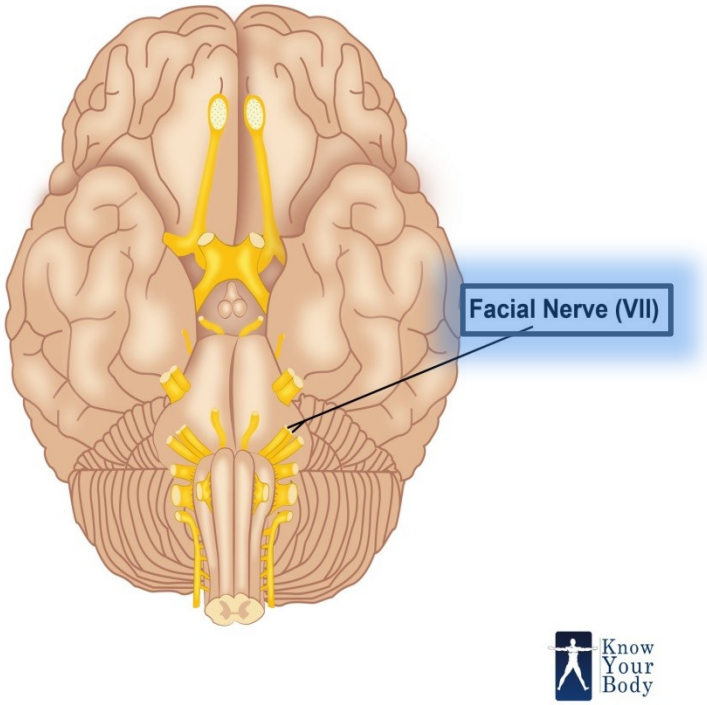
Learning Objectives:

At the end of the lecture, the students should be able to:

- **Define site of origin & type of facial nerve.**
- **Enlist nuclei & main branches of the nerve.**
- **Describe main course & areas of distribution of each of its branches.**
- **Define Bell's palsy.**

Introduction:

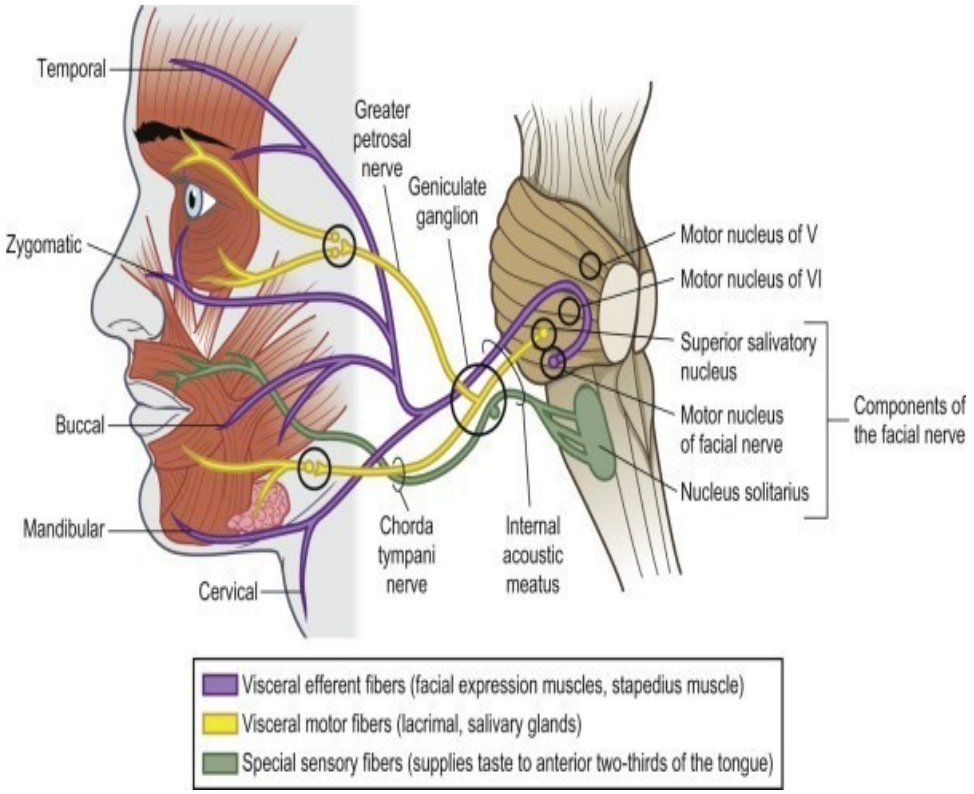
- It is seventh (VII) cranial nerve.
- It is a mixed nerve i.e. both motor & sensory.
- It also contains parasympathetic fibres.
- It is the nerve of second branchial arch.



Nuclei:

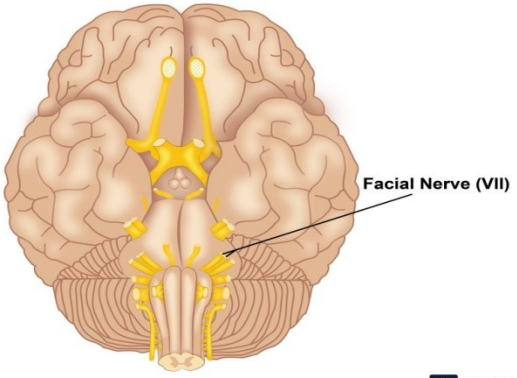
- **The facial nerve has three nuclei, viz:**

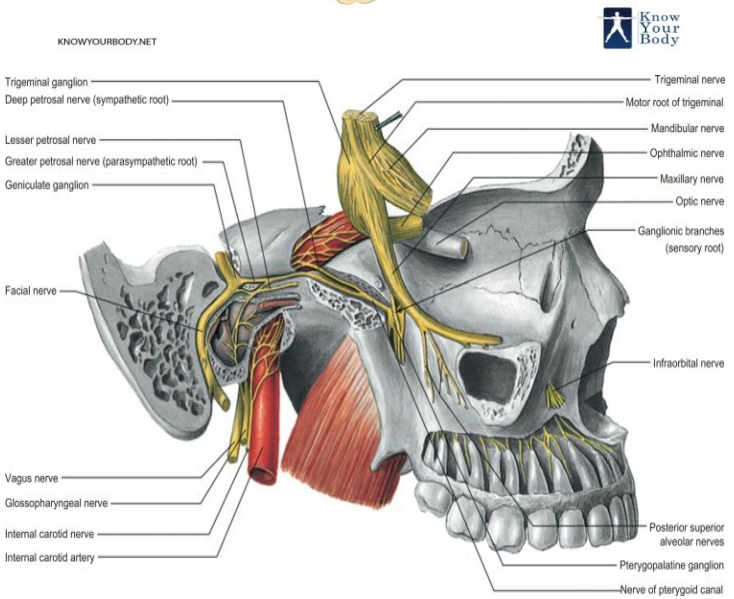
1. **The main motor nucleus**
2. **The parasympathetic (superior salivatory & lacrimal) nuclei**
3. **Nucleus of tractus solitarius**



Origin & course:

- It arises as a motor & a sensory root (nervus intermedius) anteriorly from ponto-medullary junction.
- Then run laterally & enter internal acoustic meatus in petrous temporal bone.
- At the bottom of the meatus, the motor & sensory roots fuse to form single trunk and enters the facial canal.





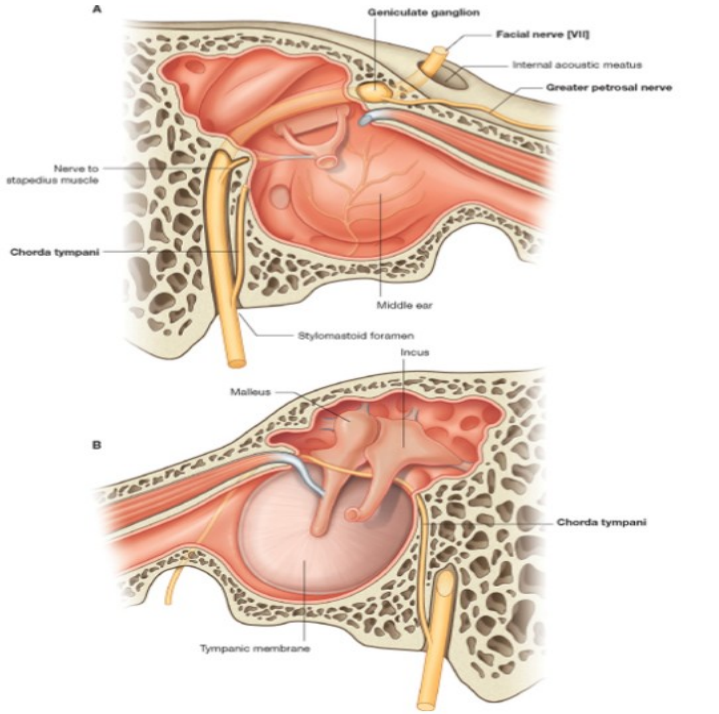
Course (continued):

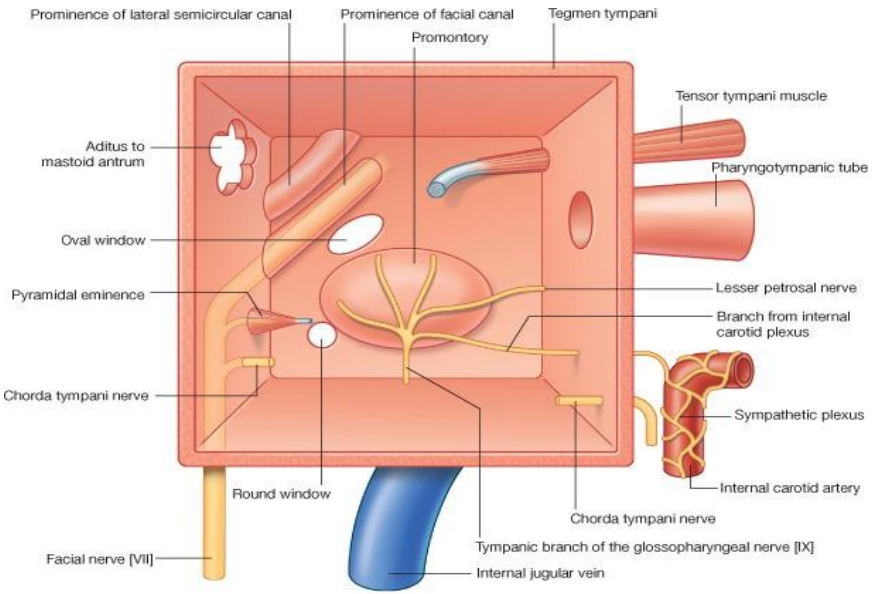
- **Here It is related directly to internal, middle & external ear and divided into 3 parts:**

1. **Labyrinthine part (internal ear)**
2. **Tympanic part (middle ear)**
3. **Mastoid/vertical part (external ear)**

- **It exits the skull through stylomastoid foramen.**

<u>Geniculate ganglion (Genu):</u> On reaching the medial wall of middle ear, the nerve expands to form sensory geniculate ganglion.

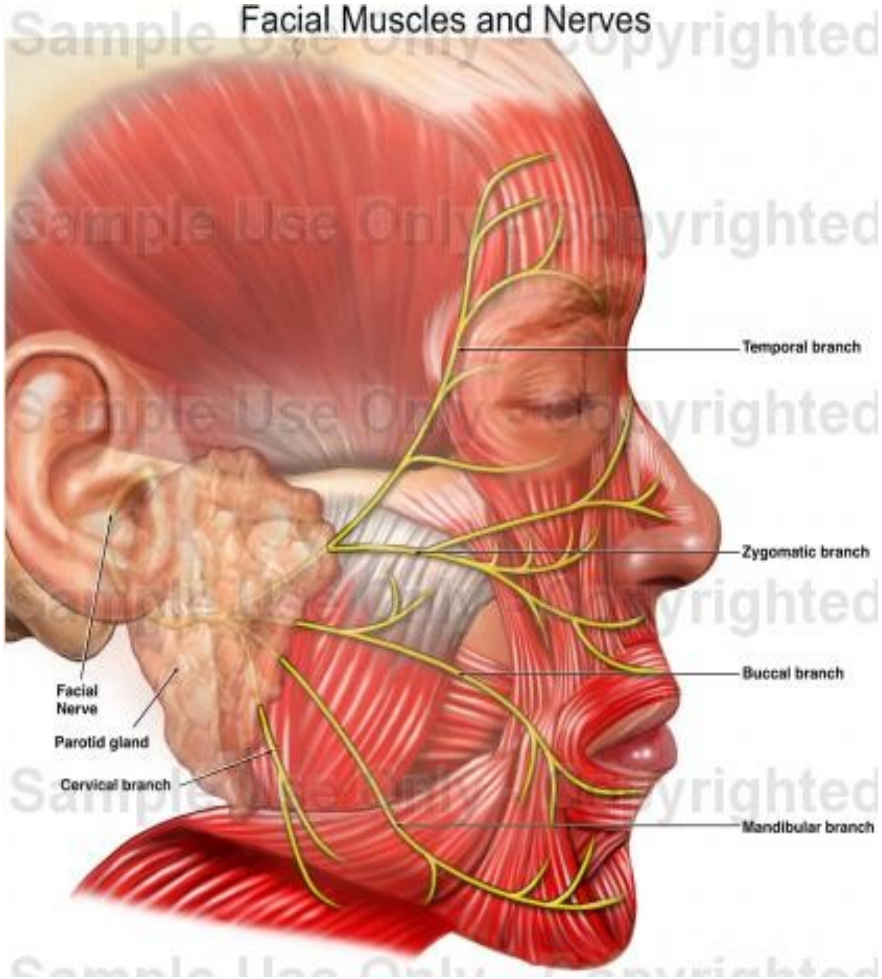




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Course (continued):

- In its extracranial course, it passes laterally to styloid process & enters posteromedial surface of parotid gland.
- Behind neck of mandible it gives off terminal branches which emerge along anterior border of parotid gland.



Branches:

A. In facial canal:

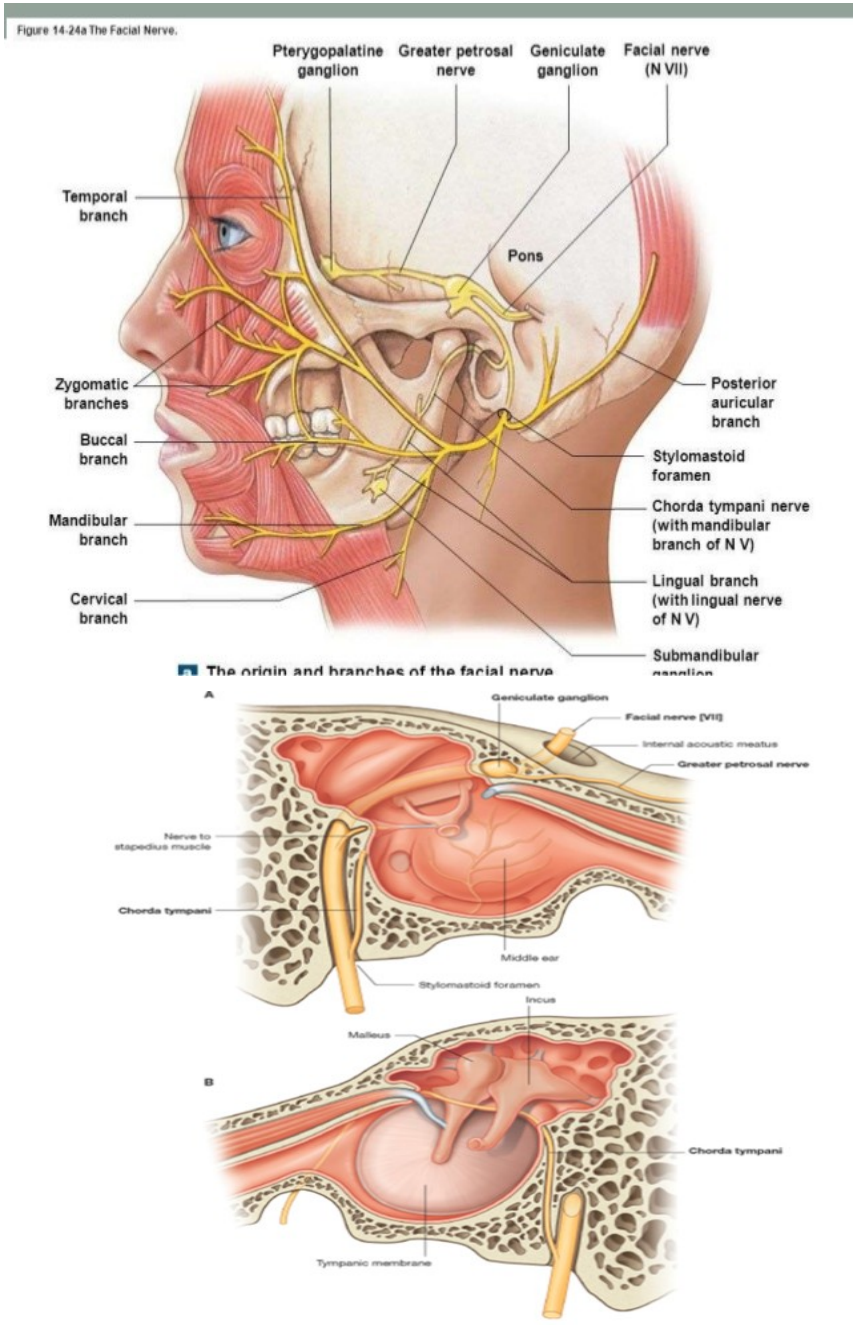
1) Greater petrosal nerve,
2) Nerve to stapedius (injury to this nerve causes Hperacusis) &
3) Chorda tympani

B. At its exit from stylomastoid foramen:

1) Posterior auricular,
2) Digastric &
3) Stylohyoid branches

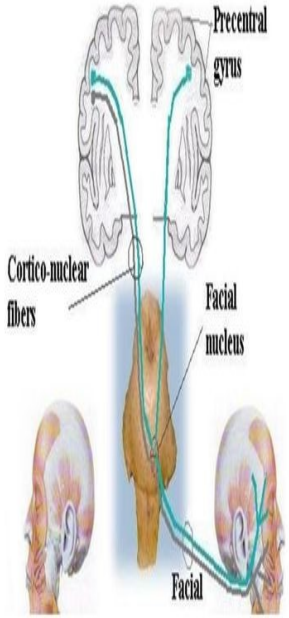
C. Terminal branches:

1) Temporal,
2) Zygomatic,
3) Buccal,
4) Marginal mandibular &
5) Cervical branches



Bell’s palsy (Lower motor neuron lesion) & Upper motor neuron lesion

Lower Motor Neuron Lesion	Upper Motor Neuron Lesion
Results from **injury of facial nerve** fibres: in internal acoustic meatus; in the middle ear; in the facial canal, or in parotid gland.	This occurs after injury to the pyramidal tract (corticonuclear) above facial nucleus...
Manifested by **complete paralysis of facial muscles** on the **same side** of lesion.	Leads to **paralysis of facial muscles of lower ½ of face of opposite side** but the upper ½ of the face not affected because the lower part of facial nucleus & Ms.of lower ½ of face receive pyramidal fibres from opposite cerebral cortex only, while Ms..of upper ½ of face are normal because they receive pyramidal fibres from both cerebral hemispheres.
If lesion of facial nerve **above the origin of chorda tympani and nerve to stapedius,** the paralysis of facial muscles will be associated with : 1- Hyperacusis : sounds are heard more acute due to paralysis of stapedius ms. 2- Loss of taste sensation from anterior 2/3 of tongue.	

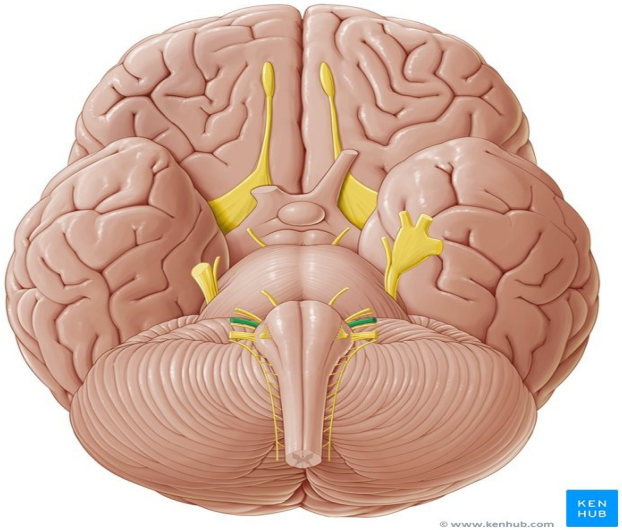


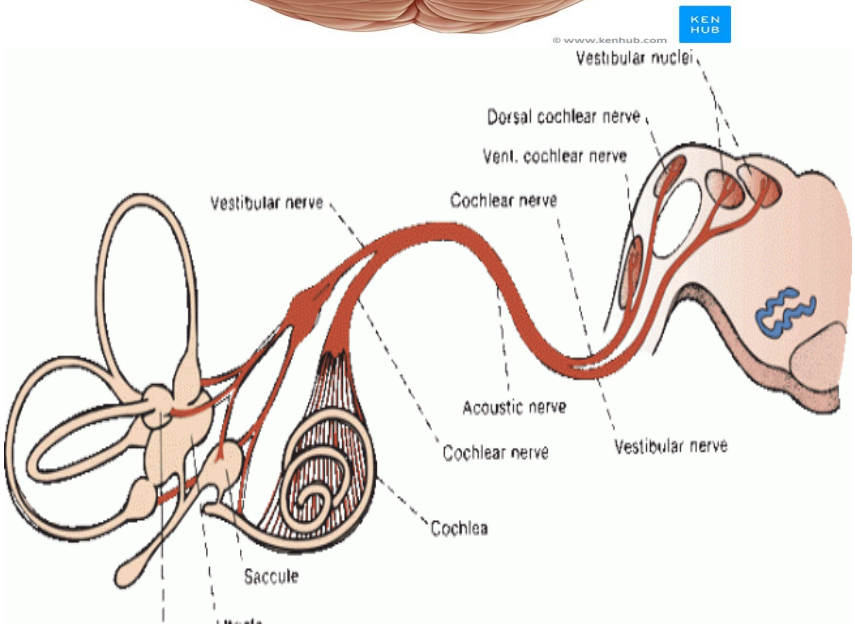


The Vestibulocochlear Nerve

Introduction:

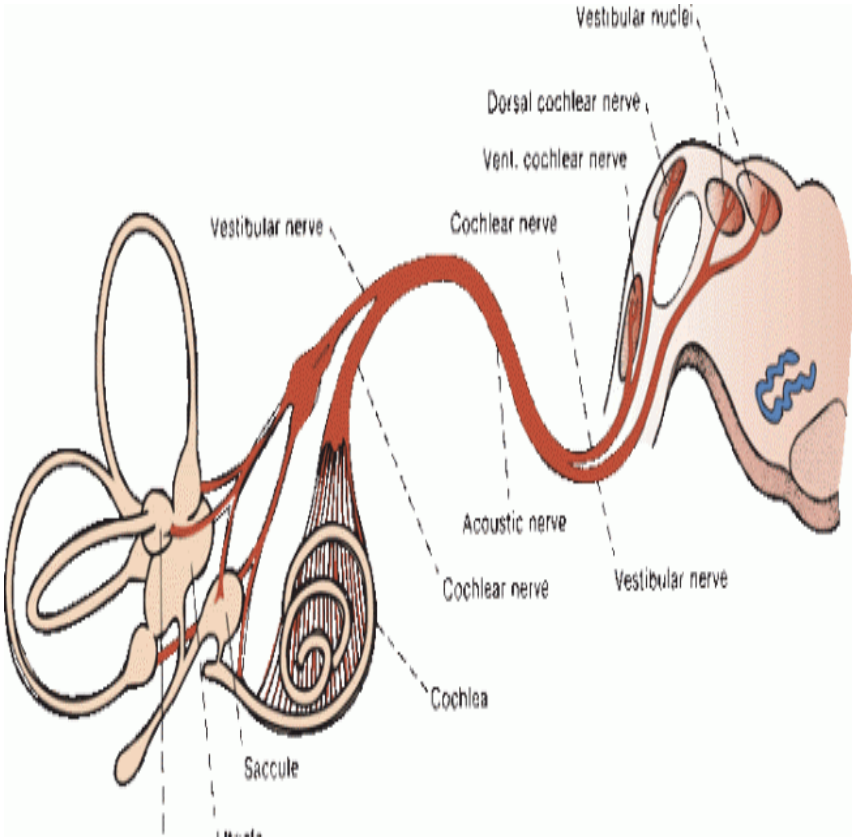
- It is eightth cranial nerve & sensory in nature.
- It has two distinct parts, the vestibular nerve & the cochlear nerve which function to transmit afferent information from internal ear to the CNS.
- Vestibular part is concerned with equilibrium & cochlear part is concerned with hearing.





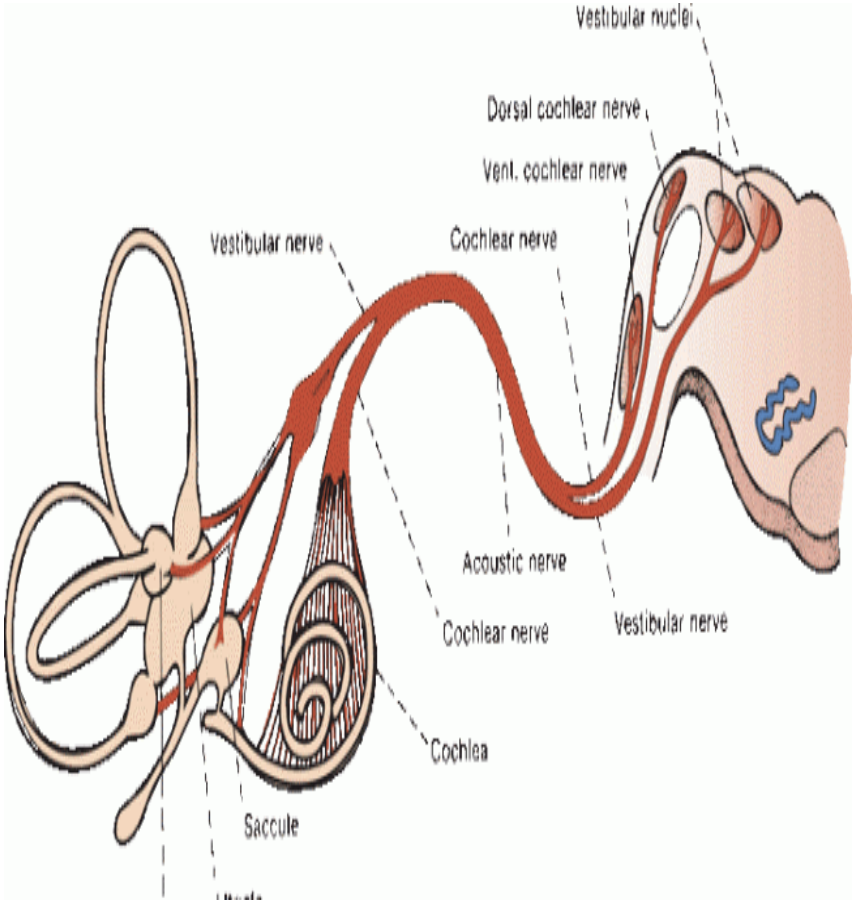
Nuclei:

- **Vestibular nuclear complex consists of four nuclei situated partly in pons & partly in medulla beneath the floor of 4th ventricle, i.e. lateral, superior, medial & inferior vestibular nuclei.**
- **Nuclei of cochlear nerve are situated in pons in relation to inferior cerebellar peduncle. They are posterior & anterior cochlear nuclei.**



Course:

- **Cochlear nerve:** **The auditory receptors are hair cells of organ of Corti located in the internal ear. They are innervated by bipolar neurons of spiral ganglion, the central processes of which form the cochlear nerve which ends in posterior & anterior cochlear nuclei.**
- **Vestibular nerve:** **The vestibular receptors for static balance are located in saccule & utricle and for kinetic balance in semicircular ducts. They are innervated by bipolar neurons of vestibular ganglion, the central processes of which form vestibular nerve which ends in vestibular nuclei.**
- **The vestibular & cochlear parts of nerve leave anterior surface of brainstem at ponto-medullary junction & enter internal acoustic meatus with facial nerve & distributed to different parts of internal ear.**




Thank You!